

Heuristic Evaluation of User Interfaces: Exploration and Evaluation

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ABSTRACT

“Heuristic evaluation of user interfaces” is a seminal work in the research and practice of human computer interaction. Widely cited, a narrative of practicality inspired an uptake in usability evaluations, stimulated the research and practice of using heuristics as a usability evaluation method. User interface heuristics were expanded on and developed for different platforms and interactions, though evaluation challenged the method’s discounting of evaluator expertise, context, dogmatic approach, and lack of reliability, validity and quantitative justification. Heuristic evaluation remains influential and is considered valid provided caveats are respected and usage is supported by quantitative data analysis and other UEMs, notably empirical user testing.

Author Keywords

Human Computer Interaction, Discount Usability, Heuristics Evaluation, User Interface Inspection

INTRODUCTION

“Heuristic evaluation of user interfaces”¹ (Nielsen and Molich 1990b) has been cited 1256 times in academic research (Source: Google Scholar), and now regarded as a usability industry standard (Spool and Schroeder 2001). Publication at the ACM CHI Conference of 1990 encouraged an uptake in software user interface (UI) usability evaluation (Cockton and Woolrych 2002), practitioners attracted by methodological reliance on a short, plain-language list of heuristics (guidelines) and a few evaluators that circumvented complexity, financial expense, execution time, technical expertise and other constraints that ordinarily preventing usability being part of the software development lifecycle.

Heuristics for a range of platforms and contexts led to the approach becoming the most frequently used usability evaluation method (UEM) (Hornbæk and Frøkjær 2004) and likely to continue so (Sauro 2004). However, practitioners and researchers are critical of the work’s research methodology and lack of quantitative statistical support.

In this paper, the work’s motivations and findings are examined, usability heuristics in human computer interaction (HCI) literature reviewed, an evaluation of the

work offered based on research and practice, and the influence of the work on HCI assessed.

HEURISTIC EVALUATION OF USER INTERFACES

Research Methodology

Usability can be usefully defined (International Organization for Standardization 1998) as:

“The effectiveness, efficiency and satisfaction with which specified users achieve specified goals in particular environments.”

In response low levels of empirical user testing or other forms of usability evaluations being practices due to awareness, time, cost and expertise constraints, Nielsen and Molich (1990b) propose the use of a heuristic-based method as a practical and efficient alternative.

The authors analyze problems found in four usability evaluations of UIs, concluding that aggregated results of heuristic based evaluation are more effective at finding usability problems than individually performed heuristic evaluation.

Before UI evaluation, the authors establish a list of known problems. Evaluators are instructed in nine usability heuristics “generally recognized in the user interface community” (Nielsen and Molich 1990b, p. 250): simple and natural dialogue, speak the users language, minimize user memory load, be consistent, provide feedback, provide clearly marked exits, provide shortcuts, good error messages, and prevent errors.

Evaluators review each UI and attempt to find as many usability problems as possible. Reporting methodology is a written report submitted by each evaluator for review by the authors. Each evaluator works in isolation and does not influence another’s findings. The authors determine whether each usability problem found is, in their opinion, a problem, score each problem identified using a liberal method. The UIs are not subsequently subjected to empirical user testing or other UEMs.

Evaluation of Four User Interfaces

The four UIs are:

Teledata, a printed set of screen captures from a video text-based search system. The evaluators are 37 computer science students from a UI design class. There are 52 known usability issues.

Mantel, a screen capture and written specification used to

¹ Nielsen, J. and Molich, R, 1990. Heuristic evaluation of user interfaces. *Proceedings of the ACM CHI 90 Conference*, 249-256.

² As of 4 December 2011.

search for telephone subscriber details. The evaluators are 77 readers of the Danish *Computerworld* magazine, the test originally run as a contest for financial reward (Nielsen and Molich 1990a). There are 30 known usability issues.

Savings, a live system, is a voice-response system used by banking customers to retrieve financial information. The evaluators are 34 computer science students who had taken a course in UI design. There are 48 known usability issues.

Transport, also a live voice response system, used to access information about bus routes and inspected by the same set of evaluators (34) used in the Savings evaluation. There are 34 known usability issues.

Evaluation Findings

The results of the individual evaluations are shown in table 1. The averages of problems found range from 20 percent to 51 percent.

UI	Number of Evaluators	Total Known Usability Problems	Average Problems Found
Teledata	37	52	51%
Mantel	77	30	38%
Savings	34	48	26%
Transport	34	34	20%

Table 1: Average individual evaluator problems found in each UI

Hypothetical aggregations using a Monte Carlo method of random sampling of between five and nine thousand sets of aggregates, with replacement, of the individual evaluators findings are then calculated. The average usability problems found by different sized groups of evaluators allow the authors to conclude that more problems are found by aggregation than by individual evaluation. The number of usability problems found increases with two to five evaluators, beginning to reach diminishing returns at 10 evaluators (see table 2). The authors say:

“In general, we would expect aggregates of five evaluators to find about two thirds of the usability problems which is really quite good for an informal and inexpensive technique like heuristic evaluation.” (Nielsen and Molich, 1990b, p. 255)

UI	Aggregates of Average Problems Found By Number of Evaluators				
	1	2	3	5	10
Teledata	51%	71%	81%	90%	97%
Mantel	38%	52%	60%	70%	83%
Savings	26%	41%	50%	63%	78%
Transport	20%	33%	42%	55%	71%

Table 2: Average individual usability problems found in each UI

For this hypothetical aggregation outcome to be realized the authors insist that evaluations be performed individually, and evaluators then jointly achieve consensus on what is a usability problem (or not) by way of the perfect authority of another usability expert or the group itself.

Core Contribution

The authors conclude that individual heuristic evaluation of UI is a difficult task for individuals, but aggregated heuristic evaluation is much better at finding problems, prescribing between three and five evaluators for a UI inspection. Advantages of this UEM are that it is inexpensive, intuitive, easy to motivate evaluators, requires no planning, and can be used early in the product development cycle. A passing recognition by the authors that evaluator mindset may influence this UEM and that it does not provide a source of design innovation, or solutions to any problems found, is made.

LITERATURE REVIEW

The literature generally focuses on evaluation the efficacy of heuristic evaluation compared to other UEMs.

Supportive

Supportive literature focuses on the cost-effectiveness of aggregated heuristics evaluation for finding usability problems. Virzi (1992) demonstrate that four or five evaluators found 80 percent of usability problems early in the evaluation cycle, a position persistently supported by Nielsen (2000). Nielsen and Phillips (1993) in a comparison of heuristic evaluation and other UEMs conclude that aggregated heuristic testing had operational and cost effectiveness advantages over others.

Supportive research generally emphasises effectiveness of the UEM when nuanced by other factors, especially evaluator expertise, and by usage early in UI development. Desurvire *et al.* (1992) show that heuristics evaluation is effective in finding more issues than other UEMs provided the inspectors are expert. Nielsen (1992) demonstrates how aggregated heuristic evaluation found significantly more major usability problems than other UEMs, and reduces the numbers of evaluators required to two or three when they have domain expertise. Nielsen. Kantner and Rosenbaum's (1997) comparison of usability studies of web sites reveals how heuristics inspection greatly increases the value of user testing later, and also acknowledges the constraints of evaluator expertise.

Wixon *et al.* (1994) show that heuristics evaluation is a cost effective way of detecting problems early enough for designers and developers to commit to fix. Sawyer *et al.* (1996) concur on commitment from product development to fix problems identified. Karat (1994) concludes that heuristics evaluation is appropriate for cost-effectiveness, organizational acceptance, reliability, and deciding on lower-level design tradeoffs.

Fu *et al.* (2002) note that heuristic evaluation and user testing together is the most effective methods in identifying usability problems. Tang *et al.* (2006) show how heuristics evaluation can find usability problems but user testing disclosed further problems.

Criticism

Work by Jeffries *et al.* (1991) conclude that although

heuristic evaluation finds more problems at a lower cost than other UEMs, it also uncovers a larger number of once-off and low-priority problems (for example, inconsistent placement of similar information in different UI screens). User testing is superior in detecting serious, recurring problems, and avoiding false positives, although the most expensive UEM to perform. Encountering false positives with heuristics is a pervasive problem, with Cockton and Woolrych (2001) showing how half of the problems detected fell into this category and Frøkjær and Lárusdóttir (1999) also reporting that minor problems are mostly uncovered. Jeffries and Desurvire (1992) also found that serious issues for real users might be missed, whereas false alarms are reported.

Finding sufficient evaluators with the expertise to use the heuristics technique is also a recurring criticism (Jeffries and Desurvire 1992). Cockton and Woolrych (2001) further probe evaluator expertise requirements positing that heuristics evaluation is more a reflection of the skills of evaluators using *a priori* usability insight than the appropriateness of the heuristics themselves.

Ling and Salvendy (2007), studying heuristics applied to e-commerce sites using a Taguchi quality control method, report that the set of heuristics impacted effectiveness because “heuristics are used to inspire the evaluators and shape what evaluators see during the evaluation”. Cockton and Woolrych’s (2009) further work reveals an instrumented impact of how 69 percent of usability predictions made were based on applying the wrong heuristics from a list.

Muller *et al.* (1995) observe that heuristics evaluation was a self-contained system of objects where contextualization of use was absent. Cockton and Woolrych (2001) expand on this with a comprehensive criticism of the method’s applicability to practice. Arguing that the real determinant of appropriateness is not the ease in which the UEM can be executed but the overall cost-benefit of the results, they declare heuristics error prone and risky, with a focus on finding problems rather than causes, while disregarding context of use, or real user impact.

Heuristic evaluation avoids the experimental controls that confidently establish causation of real usability problems. Removing expertise of user and context of use from the experiment means that false positives are reported while complex interactions (for example, completing a number of steps of actions or tasks) that might reveal critical usability errors in real usage are absent. Heuristic evaluation, then, is not encouraging of a rich or comprehensive view of user interaction.

Po *et al.* (2004) demonstrate the constraint of scenario of use and context on mobile applications evaluations, with UEMs reflective of the mobile context of use discovering more critical usability issues than heuristic evaluation, (for example, ambient lighting impact on mobile phones).

Bertini *et al.* (2006) recognize the impact of expertise and contextual factors and used Nielsen’s heuristics (1993) to derive a set reflective of mobile usage (for example, privacy and social conventions, minimalist design, and personalization). While still retaining the cost-effectiveness and flexibility of the heuristics approach, these new heuristics perform better in identifying flaws, identifying an even distribution of usability issues. Sauro (2011) recommends a combination of heuristic evaluation and cognitive walkthrough methods to redress such contextualization impacts.

The literature is deeply critical of the author’s research methodology and thus claims. Gray and Salzman (1998) are critical of the lack of sustainable inferences, generalizations about usability findings, and the cause and effects of problems, appealing for care when interpreting heuristic evaluation prescriptions. Sauro (2004) cautions use of the heuristics approach and that cost-savings are short term. Citing value when used with other UEMs, generally, heuristic evaluations shortcomings are a pervasiveness of missed critical problems, false positives, reliance on subjective opinion, and evaluator expertise requirement. Sauro (2004) decries a general HCI practitioner disdain for statistical rigor, calling for redress with quantitative data and analysis, rationales offered for variances, and provision of probability and confidence intervals as evidence of effectiveness instead of discount qualitative methodology.

The lack of common reporting formats from the UEM is an obstacle to generalized prescription (Cockton and Woolrych 2001). A requirement for agreement on a master list of usability problems, a lack of documented severity categorization and priority, and subjectivity in reporting reduces the UEM’s experimental reliability and validity (Laverty *et al.* 1997).

Expansion

The demand for simple and easily understood design guidance (Nielsen and Molich 1990a) and refactoring of usability issues (Nielsen 1994a) led to a tenth heuristic. Help and documentation (Nielsen 1994b) was added to the set that remains current at time of this paper in *Usability Engineering* (Nielsen 1993), widely available (Nielsen 2005). The original usability heuristics influenced many other acknowledged experts in the HCI field to create variants, such as the golden rules of UI design (Shneiderman 1998).

Weinschenk and Barker (2000) in the most comprehensive community of practice analysis of available heuristics across domains and platforms propose a broadly applicable set of 20 heuristics, including cultural and accessibility considerations. Kamper’s (2002) refactoring proposes 18 heuristics categorized in six groups of three overarching principles applicable across context, technologies, and domains, and is facilitative of common reporting of usability problems.

The authors' heuristics were oriented towards windows, icons, menu, and pointer-based UIs, but research led to adaptation for new user experiences, while referencing other disciplines. Hornbæk and Frøkjær's (2008) inspection technique, for example, based on metaphors of human thinking is more effective in discovering serious usability issues than regular heuristics evaluation. Reflecting the author's impact on practice, heuristics are now available for general interaction design (Tognazzini 2001), rich internet applications (Scott and Neill 2009), E-Commerce (Nielsen *et al.* 2000), groupware (Pinelle and Gutwin 2002), mobile computing (Pascoe *et al.* 2000), gaming (Korhonen and Koivisto 2006), search systems (Rosenfeld 2004), social networking (Hart *et al.* 2008), documentation (Kantner *et al.* 2002), and more.

Summary of Literature

The literature indicates that within HCI research and practice, heuristic evaluation is considered effective when supported by other UEMs, ultimately empirical user testing. Practitioners must be aware of serious constraints of context of use, evaluator expertise, and rely on tailored heuristics. False positives and missed major errors are a serious shortcoming. The literature is deeply critical of the reliability and validity of the research methodology, and lack of supporting predictability or confidence interval data leads to calls for more quantitative methodologies are brought into play. Wixon (2003) goes further; declaring that literature supportive of the UEM is "fundamentally flawed by its lack of relevance to applied usability work." (p. 34) It would appear the efficacy of heuristics evaluation, as a UEM in its own right is to iteratively uncover usability problems earlier in a development cycle when they can be fixed more easily.

EVALUATION OF THE WORK

An examination of Nielsen and Molich (1990b) against major themes emerging from research and practice reveals concerns of validity (*i.e.*, that problems found with the UEM constitute real problems for real users) and reliability (*i.e.*, replication of the same findings by different evaluators using the same test). These concerns are not necessarily ameliorated by claims, unsupported by quantitative data, that finding some usability errors is better than none at all or alluding to a vague potential evaluator mindset impact, while being symptomatic of UEM dogma (Hornbæk 2010).

Critique on quantitative data analysis grounds from Cockton and Woolrych (2001) and Sauro (2004) is particularly apt. The absence of the contextual impact, critical in usability studies, remains a central problem, and Hertzum and Jacobson (2001) point to a very significant individual evaluator effect evident, an effect restricted to neither novice nor expert evaluators, range of problem severity, or complexity of systems inspected. Molich *et al.* (2004) analysis of nine independent teams using the UEM found an evaluator effect of 75 percent of problems uniquely reported.

The authors' individual evaluator analysis demonstrates an evaluator effect (see table 3) in the minimum and maximum percentage of usability errors found by evaluators, and the variance, with some UIs appearing to be more difficult to evaluate. An explanation of the lower performing Savings and Transport voice-response systems evaluations might be offered by a low persistence of problems found (*i.e.*, an immediate response to an evaluators voice input) however, examination of the same evaluators performance on similar UI shows a weak performance correlation ($R^2=0.33$). It is suggested this performance inconsistency is due to other factors. Although the authors provided quartile and decile information, variances are not adequately explained. Qualitative methodologies such as time-on-task measurement, task completion rates, errors, satisfaction scales and asking users to complete tasks as normal would that reveal variability in evaluations are not performed.

UI	Number of Evaluators	Min %	Max %	D1 %	D9%	Q1%	Q3%
Teledata	37	22.6	74.5	26.6	67.9	43.2	58.5
Mantel	77	0 [6.7] ³	63.3	23.3	53.3	30	46.7
Savings	34	10.4	52.1	14.4	39.8	18.8	13.3
Transport	34	6.7	46.1	8.8	11.8	11.8	26.5
Average		13.2	59.3	18.3	49.2	26	40.8

Table 3: Minimum and maximum percentages of problems found by individual evaluators, along with decile and quartile analysis.

The aggregated sets of evaluations do not provide support for a Guttman scale-based hypothesis that evaluators will cumulatively find simple as well as difficult usability problems. Presented evidence is that poor evaluators can find difficult problems and good evaluators miss simple ones. The authors are dismissive of the expertise of evaluators and context when they declare:

"There is a marked difference between actual and alleged knowledge of the elements of user friendly dialogues. The strength of our survey is that it demonstrates *actual* knowledge (of usability)." (Nielsen and Molich 1990a, p. 340)

Context is a critical aspect of usage, and ability for a UEM to find a serious issue has critical validity consequences. E-commerce website near misses, for example, are a fatal usability issue, resulting in abandoned shopping carts and lost transactions (Cockton and Woolrych 2001). Analysis of the Mantel study (Nielsen and Molich 1990a) shows that the average number of serious usability problems by evaluators was 44 percent.

The authors also provide no insight into false positives, instead declaring that in their experience any given false positive is not found by more than one evaluator, with group consensus that it is not a significant problem easily achieved, while adding that "an empirical test could serve

³ The authors explain that the first evaluator found no problems. The second evaluator's findings are used.

as the ultimate arbiter” (Nielsen and Molich 1990b, p. 254). Sauro’s (2004) critique of these Type I (missed problems) and Type II (false positive) usability problem shows that without qualitative qualifiers, especially with small samples, variability and risks in usability evaluations cannot be effectively managed for real usage.

The hypothetical aggregation method, where averages of problems found are calculated using a Monte Carlo technique of random sampling (with replacement) of between five and nine thousand aggregates from the original data set of evaluators with limited usability expertise, rather than a normal distribution of evaluators undermines any claims for practical heuristic evaluation or for reliability of the claims made.

The related dependency on a perfect authority to deliver consensus and eliminate false positives or missed serious errors is left unexplored. Discussion of team dynamics or other factors that impact collective decision-making teams are outside the scope of this paper, but achieving of such consensus is not straightforward and such a critical variable requires investigation.

Hornbæk (2010) provides a useful structure to further critique, based on UEM dogma of problem counting and matching. Counting problems as a measure of potential usability issues presents difficulty from a validity perspective as it includes problems that may not be usability problems found in empirical testing or real use. Evaluators may also find problems that do not match the heuristics or the known problem list, reflected by the author’s acknowledgement that their list of problems was adjusted as evaluators found problems that were not identified by their own expertise (examples are not provided). A primacy of finding issues over prescriptions of how to fix them, or analysis of their causes in isolation of the design process, brings the validity of the UEM into question, Hornbæk (2010) concluding that:

“Identifying and listing problems remains an incomplete attainment of the goal of evaluation methods.” p. 98

Related to the counting problem is that of matching these issues to the heuristics promulgated. No information is provided on the authors matching procedure, the interpretations of what is a problem compounded by a lack of common reporting of the issues, and the reported liberal scoring. No explanation offered for the heuristics list other than they are considered by the authors to be generally recognized by the relevant practitioners as “obvious” or the authors own personal experience (Nielsen and Molich 1990b) exposes the work to further question on validity grounds.

Individual problems as a unit of usability analysis may not be reliable or practical either. Jeffries (1994) is especially critical of this assumption when he says that UEMs must:

“Ensure that the individual problem reports are not based on misunderstanding of the application, that they don't contradict

each other, that the full impact of any trade-offs are taken into account and that the recommendations are applied broadly, ...not just to the one the evaluator noticed.” (p. 290)

Cockton and Woolrych (2002) concur. A casual reading of the heuristics for good error messages, preventing errors, and use of plain language reveals empirical contradiction and overlap, for example. The heuristics and known usability problems in the authors’ study are all accorded the same weight.

Nielsen (1995) readily describes evaluation of interfaces using discount methods (of which heuristics evaluation is one) as:

“Deliberately informal, and rely less on statistics and more on the interface engineer's ability to observe users and interpret results.” (p. 98)

Yet, that the authors do not report probability of usability problems, confidence intervals of incidence of problems found, rely on subjective recommendation from a small number of evaluators where expertise and context is a critical factor, and use a qualitative (and indeed non-standard) method of reporting cannot be dismissed easily given the empirical consequences. By way of example, Spool and Schroeder (2001) challenge the industry standard claims about five evaluators finding 85 percent of errors as invalid, citing the impact of product, investigators, and techniques when five evaluators found 35 percent of known problems. Gray and Salzman (1998) are also critical of the validity of the experiments, and Cockton and Woolrych (2002) call attention to the small number of evaluators. Sauro (2004) and Virzi's (1992) use of the formula $1-(1-p)^n$ to estimate the sample sizes needed to predict probability of a problem being found shows that more than five users are required⁴ if probability and confidence intervals are to be managed and validity assured. Sauro (2004) recommends that practitioners understand the risks involved in heuristic evaluation and use a combination of UEMs, gathering both quantitative and qualitative data, adds:

“If you accept the prevailing (ISO) definition of usability, you must also accept that measuring usability requires measures of effectiveness, efficiency, and satisfaction—measures that move you into the realm of quantitative methods”. (p. 34)

INFLUENCE AND CONCLUSION

Nielsen and Molich (1990b), inspired an uptake in usability practice and a thriving debate about the relative effectiveness of empirical usability testing versus what has entered HCI parlance as discounted UEMs (Nielsen 1994). As a result, heuristic evaluation eased industry uptake of HCI methods in the 1990s (Cockton and Woolrych 2002), and became the most widely used UEM in practice

⁴ Virzi (1992) shows how for a 90 percent confidence level, 22 users would be needed to detect a problem experienced by 10 percent. The formula used is $1-(1-p)^n$, where p is the mean probability of detecting a problem and n is the number of test subjects.

(Hornbæk and Frøkjær 2004).

Although Nielsen (1995, 2004) consistently argues that even without the power of statistics, some usability testing, performed iteratively, and the finding some problems is better than none at all, particularly for interfaces still to be implemented, the reliability and validity of those claims indicate extreme caution for practice. Cockton and Woolrych (2002) declare that (such UEMs):

“Rarely lead analysts to “consider how system, user, and task attributes will interact to either avoid or guarantee the emergence of a usability problem.” (p. 15)

Cockton and Woolrych (2001) acknowledge that heuristic evaluation has a place driving design iterations and in increasing usability awareness, but understanding limitations of context of use, total cost, and how to mitigate constraints is critical for practice. Spool and Schroeder (2001) recognize there is validity to the method provided an understanding of the numbers of evaluators is required as well as constraints of features, individuals testing techniques, the complexity of task, and nature or severity of the problem. They insist the author’s rule of thumb approach to number of evaluators must be countered by quantitative approaches and supplemented by other methods.

The effective contribution of heuristic evaluation can be maximized by operational considerations, with iterative inspections made early on in UI development, identifying more obvious lower performance issues, thus freeing resources to identify higher-level issues with real user testing. However, there is no one single best UEM and the search for one is unhelpful for practice (Hornbæk 2010). Usability practitioners use, and will continue to use, a combination of methods. Hollingsed and Novick (2007) concur that empirical and inspection methods are widely used together, a choice made on the basis of what is most appropriate for the context and purpose of evaluation. Fu *et al.* (2002) show that users and experts find fairly distinct sets of usability problems, and summarize that:

“To find the maximum number of usability problems, both user testing and heuristic evaluation methods should be used within the iterative software design process.” (p. 142)

Heuristics evaluation has its place for easily finding low-hanging fruit problems (of various severities) early in design cycle, and continues to offer value as a UEM. As practitioners become aware of the limitations of the method and become adept at understanding the implications of UEM choice decisions the risks of usability heuristics as a standalone methodology become less significant.

Notwithstanding that user testing remains the benchmark for usability evaluation, that heuristics have emerged for web-based, mobile and other interactions serves as testament to the enduring seminal nature of the authors’ work. Although models of rapidly iterative and shorter innovation cycles, agile-based software development and

ad hoc or cloud-based testing scenarios and emergent new interactions (mobile, gamification, augmented reality, and so on) are beyond the scope of this paper, their prescience and now accepted acknowledgement of the importance of usability in UI development, means that research into heuristic evaluation and its practice will continue.

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